PRODUCT REQUIREMENT DOCUMENT

PRD: Compliance QA Simulation Tool

PRD-2026 · AIAvenue Agency

Executive Summary

The Compliance QA Simulation Tool is a self-contained, zero-dependency browser-based prototype that simulates automated insurance compliance scanning. Built entirely with vanilla HTML, CSS, and JavaScript, the tool opens in any browser without a server, installation, or build step. It offers a realistic compliance operations dashboard featuring real-time statistics, a dynamic checklist builder with toggleable rule categories (Color, Text, Legal/IRDAI, Image, Brand), a simulated compliance scan with animated progress, a detailed violation breakdown interface, a maker-checker two-person approval workflow, and a fully filterable audit log. The prototype is themed via CSS custom properties and uses Tabler Icons for visual elements. It is designed for product demos, stakeholder walkthroughs, and UX testing before committing to a full-scale engineering build — targeting compliance officers, QA managers, and insurance operations teams who need to evaluate automated compliance scanning workflows.

Problem Statement

Core Problem

Insurance companies in India must ensure every customer-facing document — policy wordings, claim forms, marketing brochures, agent scripts — complies with IRDAI regulations, brand guidelines, and legal standards. Manual compliance checks are slow, inconsistent, and error-prone: a single policy document can take 45–90 minutes to review, and human reviewers miss 12–18% of violations on average (industry internal audit data).

User Pain Points

- **Manual Review Bottleneck:** A team of 5 compliance officers reviews ~80 documents/day. Each document takes 30–90 minutes. Backlogs routinely exceed 48 hours — directly delaying product launches and claim payouts.
- **Inconsistent Standards:** Two compliance officers reviewing the same document may flag different issues. There is no machine-verifiable standard for what constitutes a violation in colour contrast, font size, or legal disclaimer placement.
- **No Rule Reusability:** Compliance rules are stored in Excel sheets, Slack messages, and individual reviewers' notes. When IRDAI releases new guidelines, it takes 2–3 weeks to propagate changes across the team.
- **Audit Trail is Manual:** When a non-compliant document reaches a customer and a regulator asks who approved it, the answer is "I don't remember" — there is no systematic audit log.
- **Maker-Checker is Circumvented:** The two-person approval workflow (maker drafts → checker approves) exists on paper but in practice, reviewers approve documents without thorough re-checks due to time pressure.
- **No Quantified Compliance Score:** Teams cannot answer "How compliant are we overall this month?" — there is no aggregate compliance health metric across all document types.

Why a Prototype First?

- Stakeholders need to **see and interact** with automated compliance scanning before committing to a full engineering build
- Compliance rules vary significantly across insurers — a prototype enables rapid rule set iteration
- No backend dependency means the prototype can be shared via email attachment or USB drive

- UX patterns can be validated with real compliance officers in 1-hour testing sessions

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User Research & Personas

Persona 1: Kavita Reddy — Compliance Officer

Attribute	Detail
Age	36
Role	Senior Compliance Officer at a mid-sized life insurer
Tech Comfort	Moderate — uses Excel, PDF readers, internal portals
Daily Workflow	Opens 12–15 PDFs per day; checks each against a printed checklist; marks violations in an Excel tracker; sends approval email; logs PDF to shared drive
Pain Points	Carpal tunnel from repetitive checks; misses violations when tired; Excel tracker is never up to date
Needs	Automated scanning that flags violations instantly; visual dashboard showing compliance health; simple checklist builder for rule management

Persona 1: Kavita Reddy — Compliance Officer

Persona 2: Rajan Gupta — QA Manager

Persona 3: Meera Iyengar — VP Operations

Market Analysis

Total Addressable Market (TAM)

The global insurance compliance software market was valued at **$4.8 billion in 2024**, projected to reach **$8.2 billion by 2029** (CAGR 11.3%). India's insurance market, growing at 15% CAGR (IRDAI annual report), represents approximately **3–4%** of the global compliance software spend, yielding a TAM of **₹1,200–1,600 crore**.

Serviceable Addressable Market (SAM)

Indian insurers requiring document compliance automation: **56 life insurers, 34 general insurers, 6 standalone health insurers, and ~120 corporate agents/brokers** (IRDAI registered). Average compliance department size: 8–15 people. At an estimated software spend of ₹8–15 lakh per insurer per year for compliance QA tools, the SAM is approximately **₹173–324 crore/year**.

Serviceable Obtainable Market (SOM)

Targeting the prototype's buyer persona (operations heads at mid-sized insurers who want to pilot automation before committing to enterprise tools): ~45 insurers in the ₹500–2,000 Cr premium range. At a prototype-to-pilot conversion rate of 20% and pilot-to-full-deployment of 60%, plus ₹15 Lakh average deal size, the SOM is **₹8.1 crore in Year 1** (27 pilots × ₹30L = ₹8.1 Cr) and **₹32.4 crore by Year 3** (with full deployment to 54 insurers at an average ₹6 L/yr SaaS).

Market Rationale

- IRDAI mandated stricter disclosure norms in 2024 (Master Circular on Product Literature) — creating urgency for compliance automation
- India's insurance penetration is 4.2% and growing — more documents, more scrutiny
- No Indian-market-specific compliance scanning tool exists (most are US/UK-centric, do not understand IRDAI guidelines)
- A zero-dependency prototype dramatically reduces sales friction — compliance teams can test without IT involvement

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Competitive Analysis

Competitor	Strengths	Weaknesses	Our Edge
Zoura / Compliance.ai	Strong US regulatory focus, ML-based scanning, enterprise SaaS	US-centric (SEC/FINRA), not IRDAI-aware, expensive (₹2 Cr+/year), heavy integration	IRDAI-native rule library, India pricing (₹6 L/yr), no server setup, prototype-first sales motion
Manual Excel + Checklists	Zero cost, highly customisable, no training needed	70% slower, 18% miss rate, no audit trail, no real-time dashboard	Automated scanning in seconds, quantified compliance score, filterable audit log
Thomson Reuters ONESOURCE	Global regulatory coverage, tax + compliance combo	Overkill for document compliance only, extremely complex, 6-month implementation	Purpose-built for insurance document QA, 1-hour setup (just open the browser), instant value
DocMetric	Claims-focused document automation, AI extraction	Not compliance-scanning focused, no IRDAI rules, no maker-checker workflow	Dedicated compliance scanning engine, rule builder, maker-checker as first-class feature
Homegrown Excel Macros	Some automation, low cost, already in use	Maintenance nightmare, breaks with Excel updates, no audit log, no real-time UI	Modern web UI, real-time dashboard, persistent audit log, visual rule builder, portable

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Solution Overview

Core Features

F1 — Real-Time Compliance Stats Dashboard

- 6 stat cards at the top: Documents Scanned, Total Violations, Critical Violations, Compliance Score (%), Open Actions, Pending Approvals
- Colour-coded KPIs (green ≥ 90%, amber 70–89%, red < 70%)
- Auto-refresh on scan completion — no manual reload needed
- Tabler Icons for each stat (File Scan, Alert Circle, Shield Check, etc.)

F2 — Checklist Builder with Toggleable Rules

- 5 rule categories, each expandable to show individual rules:

- **Color Rules:** Contrast ratio ≥ 4.5:1, no prohibited colour combinations, brand colour compliance - **Text Rules:** Minimum 12pt font size for body, heading hierarchy compliance, no all-caps passages exceeding 10 words - **Legal/IRDAI Rules:** Mandatory disclaimers present, IRDAI registration number visible, "Terms and Conditions apply" on all marketing material - **Image Rules:** All images have alt text, no stretched/logo images, image resolution ≥ 300 DPI for print - **Brand Rules:** Logo in top-left corner, brand fonts used, tagline present on cover page

- Each rule has a toggle switch (ON/OFF) — allows customisation per document type
- "Select All / Deselect All" per category
- Active rule count displayed in header: "12 of 28 rules active"

F3 — Simulated Compliance Scan

- "Run Scan" button triggers a full-document scan simulation
- Animated progress bar with status messages: "Checking colour contrast...", "Validating IRDAI disclaimers...", "Scanning brand compliance..."
- 8–12 second animation (configurable) — realistic enough for demos
- On completion: toast notification "Scan complete — 14 violations found"

- Violation count per category shown as doughnut chart

F4 — Violation Breakdown

- Table list of all detected violations with columns: Rule Name, Category, Severity (Critical/Major/Minor), Description, Auto-Fix Available?
- Severity badges with colour-coding (Critical = red, Major = amber, Minor = blue)
- Clicking a violation row expands a detail panel with: the violated rule text, excerpt from document, suggestion for fix
- Group violations by category or severity with filter buttons
- Bulk action footer: "Approve All Minor Violations" — for low-risk items

F5 — Maker-Checker Approval Workflow

- **Two-person approval model:**

- **Maker (role):** Reviews scan results, marks violations as "Fixed" or "Needs Review", submits for approval - **Checker (role):** Reviews maker's decisions, can approve or reject each; final approval locks the document

- Status badges: Pending Maker, Pending Checker, Approved, Rejected
- Timeline view showing maker action → checker action with timestamps
- "Approve All" button for checkers (with confirmation modal — "Are all violations genuinely resolved?")

F6 — Filterable Audit Log

- Searchable log of all documents scanned, results, and approval actions
- Filters: Date range, Rule Category, Severity, Document Type, Status, Assigned To
- Columns: Document Name, Scan Date, Violations, Compliance Score, Status (Approved/Rejected/Pending)
- Export to CSV button
- Pagination (20 items per page)

Architecture (Zero-Dependency)

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┌──────────────────────────────────────────────┐
│ index.html (Single File) │
│ │
│ ┌──────── CSS ────────┐ ┌── Tabler Icons ──┐ │
│ │ Custom Properties │ │ (CSS-only icons │
│ │ (theming, colours, │ │ via CDN link — │
│ │ spacing, typography) │ │ no JS dependency) │
│ │ └──────────────────────┘ └──────────────────┘ │
│ │ ┌──────── JS
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────────────────────────────────┐ │ │ • State: Plain object (no framework) │ │ │ • Rendering: Template literals + innerHTML │ │ │ • Routing: Hash-based SPA │ │ │ • Storage: localStorage for rules + audit log │ │ │ • Scan simulation: setTimeout promise chain │ │ │ • Charts: Canvas-based doughnut chart (custom) │ │ │
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CSS Custom Properties Theming

- `--color-primary`, `--color-success`, `--color-danger`, `--color-warning`
- `--font-family`, `--font-size-base`, `--border-radius`
- `--sidebar-width`, `--header-height`, `--spacing-unit`
- `Light/dark theme toggle via` [data-theme="dark"]` selector
- All components reference these variables — instant re-skinning for client demos

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Feature Deep-Dive (RICE Priority Matrix)

Feature	Reach (demo users)	Impact (1–5)	Confidence (%)	Effort (person-days)	RICE Score
F1 — Real-time stats dashboard	100% (all see it first)	5	95%	2	237.5
F2 — Checklist builder with toggles	80% (compliance officers)	4	90%	3	96.0
F3 — Simulated scan with progress	100% (demo hook)	5	95%	2	237.5
F4 — Violation breakdown table	80% (reviewers)	4	90%	3	96.0
F5 — Maker-checker workflow	60% (approval flow demos)	5	85%	4	63.8
F6 — Filterable audit log	70% (managers)	3	85%	2	89.3
CSS custom properties theming	100% (reskinning value)	3	95%	1	285.0
localStorage persistence	50% (return users)	3	90%	1	135.0
Doughnut chart (custom canvas)	70% (dashboard view)	2	85%	1.5	79.3
Export to CSV	40% (reporting need)	3	90%	0.5	216.0

Top Priorities (Prototype): 1. CSS custom properties theming (285.0) — foundational for client customisation 2. F1 + F3 — Dashboard and Scan animation (237.5 each) — demo hook 3. Export to CSV (216.0) — reporting expectation 4. localStorage persistence (135.0) — realistic UX 5. F2 + F4 — Checklist and Violations (96.0 each) — core functionality

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Success Metrics (KPIs)

KPI	Target (Prototype Phase)	Measurement
Stakeholder demo conversion	> 60% request full build	Post-demo survey
UX testing task completion	> 85% of users find all features in < 3 min	Moderated UX test (8 users)
Rule builder usability	> 90% toggle all 5 categories in < 30 sec	Session recording
Scan simulation engagement	> 80% of demo users watch full scan animation	Interaction tracking in prototype
Audit log discovery	> 70% of test users find and filter audit log	Unmoderated task completion
Prototype sharing rate	> 50% of recipients forward to > 1 colleague	Unique visit tracking (analytics pixel)
Critical bug count	0 (vanilla HTML/CSS/JS — no framework bugs)	Manual QA pass

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Roadmap

Phase 0 — Prototype Build (Week 1, 5 days)

- Single `index.html` with embedded CSS + JS
- CSS custom properties theme system with light/dark toggle
- F1 — Stats dashboard with 6 semantic stat cards (Tabler Icons)
- F3 — Simulated scan with 8–12 second animated progress
- F4 — Violation breakdown table with severity badges
- F2 — Checklist builder with toggleable rules and categories
- F5 — Maker-checker approval workflow (basic: two-state toggle)
- F6 — Filterable audit log with localStorage persistence
- Export to CSV function
- Custom canvas-based doughnut chart for violation distribution
- Responsive layout (down to 1024px)

Phase 1 — Stakeholder Validation (Week 2–3)

- Internal team demo + feedback loop
- Identify top-5 friction points
- Add 2–3 rule examples per category (realistic IRDAI rules)
- Polish micro-interactions (hover states, transitions, loading skeletons)
- Add analytics pixel (optional, for sharing metrics)
- Prepare demo script and use-case scenarios (life insurance, health insurance, general insurance)

Phase 2 — UX Testing + Iteration (Week 4–5)

- Moderated UX testing with 8 compliance professionals
- Measure task completion times and error rates
- Iterate on: stat card label clarity, scan animation speed, rule builder discoverability
- Add tooltips for compliance terms ("What is contrast ratio?")
- Add keyboard accessibility (tab navigation, aria labels)
- Fix all usability issues identified in testing

Phase 3 — Full Build Greenlight (Month 2–3)

- Present prototype + UX test results + market analysis to leadership
- Build business case for full engineering build (React/Node/Python → microservices architecture)
- Estimate full engineering effort
- Target: convert prototype into production-ready SaaS platform with:

- Real PDF/image scanning using OCR + computer vision - AI-driven rule learning - Multi-tenant SaaS deployment - Integration with insurer document management systems - Automated compliance reporting for IRDAI audits

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Impact Estimate

Impact of Prototype Alone

Metric	Value
Cost to build	₹50,000 (5 days of senior frontend developer)
Stakeholder demos conducted	15 (target)
Expected pilot conversions	9 (60% conversion)
Pilot deal size	₹30,00,000 (₹3 Lakh setup + ₹6 L/yr SaaS × 5 years)
Prototype-driven pipeline value	**₹27,00,00,000** (27 Cr from 9 pilots)
Prototype ROI	**5,400×** (₹50K cost → ₹27 Cr pipeline)

Impact of Prototype Alone

Projected Impact of Full Build (Year 3)

Financial Impact (Full Build, ₹/year)